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Factory-Correct Restoration Reference

LS/LT Conversion Guide — 1973–1987 GM Squarebody Trucks

Model Year 2026

Recipient trucks	Chevrolet/GMC C/K pickups, Blazer/Jimmy, Suburban (1973–87)
Donor engines	Gen III/IV LS truck donors · crate LS3 · Gen V LT
Highlights	Dual saddle-tank fuel plan; 4WD notes; worksheets included
Price	\$29.95 — PDF download at checkout

Every specification in this guide was checked against factory literature, marque references, period road tests, and documented survivor cars. Where sources disagree or the record is silent, this guide says so rather than guessing.

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Chapter 1 — The Decision: Donor, Crate, or Gen V LT (Read This First)

The 1973–87 GM truck — the squarebody — is the easiest LS swap in the classic-vehicle world, which is exactly why the most money gets wasted on it: 'easy' makes people skip the planning chapter. The engine bay is huge, the mounts are a solved problem, and the wrecking yards are full of donors. What still goes wrong, in order, is budget shock (the engine was the cheap part), the saddle tanks, and a truck that runs but has gauges, fans, and a speedometer living in three different decades. This chapter prices the whole job before the first bolt turns, the way the family does it for every body.

- **Donor-first (the default, and this guide's main road):** a salvage truck 5.3 or 6.0 (LM7/LQ4/LQ9 family) with its harness, PCM, and accessories. A truck donor into a truck recipient is the swap gods' gift: the accessory drive and intake were packaged for this exact engine bay shape. The unknowns are mileage, maintenance, and accessory completeness — audited in Chapter 2 before money moves.
- **Crate:** a new LS3 or a controller-package crate with its wiring and calibration. You are buying certainty about the engine and nothing else — the tanks, the pump, the cooling, the gauges are the same bill either way. Do the arithmetic in Worksheet C before romanticizing either path.
- **Gen V LT:** the direct-injected L83/L86 truck engines. Better efficiency, real low-end, and a different job: high-pressure fuel hardware, VVT/AFM behavior, and packaging that chapter 4 covers on its own. Swapping an LT like it was an LS is the expensive way to learn the difference.

The line items first-timers miss (planning ranges, not promises — salvage and parts prices move): mount plates; the transmission crossmember question; an EFI fuel system end to end — and on a dual-tank truck that means TWO pumps and a selector valve that can take LS pressure; a radiator and fan decision; the harness (reworked donor vs standalone) and the PCM edits; exhaust from the manifolds back; a driveshaft measurement; gauge-drive adapters or an electronic speedometer; and the tune. If the honest total busts the budget, the endorsed alternative is to finish the truck with the drivetrain it has — a running squarebody at a cruise night beats a torn-apart one in the garage, every time.



Note. If you only have \$X: spend it in this order — (1) mounts that keep the crank centerline honest, (2) the fuel system done once at full pressure spec, (3) cooling, (4) the PCM edits, (5) cosmetics. A swap that starts, cools, and charges on day one can be dressed up for a decade. A polished engine that boils in traffic is yard art with insurance.

Chapter 2 — What You Start With: The Squarebody Baseline

Inspect the recipient truck before buying the donor. Pass/fail criteria, not vibes:

- **Frame at the steering box:** the box mounts inside the driver's frame rail, and that mounting area cracks on these trucks. Wire-brush it and look. Cracked = repair (plates/welding by someone who welds) before more power arrives. **Go/no-go.**
- **Crossmembers and mounts:** stock clamshell frame mounts present and sound on V8 trucks; 2WD vs 4WD crossmember condition. Rusted-through = replace, not plate-over.
- **Which tanks do you actually have?** Single or dual saddle tanks; steel condition inside (drop a light in through the sender hole); sender year — 1987 sending units are the EFI swap's best friend (Chapter 6).
- **Tunnel hump:** high-hump or low-hump cab floor. Measure what you have (Worksheet B) — it decides whether a 4L80E is a bolt-in or a sheet-metal project.
- **4WD trucks:** which transfer case (NP203 full-time, NP205, NP208 by year) and its crossmember — the transmission pairing plan depends on the answer.
- **The donor audit** (same discipline as the car guides): compression or a running demonstration, oil condition, and a signed accessory inventory — alternator, starter, MAF, throttle pedal (drive-by-wire donors), PCM, harness completeness. 'Complete pullout' is a sales phrase; the checklist is the contract.

Chapter 3 — Fitment: Putting an LS in a 1973–87 GM Truck

Mounts. Adapter plates bolt the LS to the stock clamshell frame mounts, and the established kits keep the crankshaft centerline and bellhousing face where the small-block had them. That single decision is why this swap is 'easy': a transmission that stays in its stock position keeps the stock tunnel, driveshaft length, and linkage geometry. Slotted plates buy fore-aft adjustment — useful, and a temptation: every inch forward is an inch the fan, the steering, and the firewall re-litigate.

Oil pan. The truck donor pan was shaped for a truck chassis and generally clears the 2WD crossmember where car pans hang low — 'generally' is doing work in that sentence. Mock it up, turn the steering lock to lock, and measure (Worksheet B). A retrofit pan exists if yours doesn't clear; find out at mock-up, not at final torque.

- **Accessory drive:** the truck drive (alternator high on the driver's side, tall intake) fits this bay the way it fits the donor's. Low-mount truck A/C compressors clear published plate kits but limit rearward engine adjustment — decide A/C before the final position, the way you decide everything else: on paper first.
- **Transmission and crossmember:** 4L60E/4L80E go behind the LS with the crossmember repositioned; whether the tunnel clears depends on the hump you measured in Chapter 2. 6-speed manuals and the 6L80 get measured, never assumed.
- **4WD:** transfer-case adapter or donor-paired case, crossmember under it, and front driveshaft clearance at full droop — a mock-up measurement, flagged in the gotchas.
- **What does NOT need cutting:** the firewall, the inner fenders, the core support. A squarebody LS swap that needs a torch is a plan that skipped this chapter.

Chapter 4 — The Gen V LT Chapter: Direct Injection Changes the Job

The Gen V truck engines (L83 5.3, L86 6.2) bolt to the same mounts and answer to the same transmission logic as the LS. Everything downstream of the fuel door is different: direct injection means a cam-driven high-pressure pump on the engine fed by a serious in-tank supply, a vacuum pump where the brake booster expects manifold vacuum, VVT and AFM (cylinder deactivation) living in the calibration, and cooling and wiring details that are LT's own. Treat the LT as a donor-engine purchase with a factory control package or a mature standalone: the lift-pump supply must hold the DI pump's inlet spec at full load, the harness match (engine/body years) matters more, and the steam vents from the heads still get plumbed — DI did not retire physics. If that paragraph reads like a project inside the project, it is: price it in Worksheet C, and let Chapter 1's decision be a real decision.

Chapter 5 — Wiring and Controls

Harness strategy — the same three roads as the car guides: reworked donor harness (cheapest, demands care and hours), standalone swap harness (the middle road most squarebody builds take), or a controller package with a crate engine. Whichever road: the PCM needs its anti-theft (VATS/passkey) edit before first start or the engine will crank forever in silence; rear-O₂ and emissions edits follow your jurisdiction's rules (stated, never guaranteed); drive-by-wire donors bring their matching pedal, mounted at sane geometry; and the fan outputs get programmed — truck PCMs don't drive electric fans by default, so the conversion is a calibration item, not a wire.

- **Gauges:** the 1973–87 cluster is electric and plays along: oil and temperature senders thread-adapt or get the donor's senders with adapters; the tach signal scales by cylinder-count output in the PCM edit. The honest problem is the speedometer: a mechanical cable drive meets a vehicle-speed sensor that only speaks to the PCM. The three honest fixes: electronic speedometer, a cable-drive conversion box, or a GPS speedometer — chosen to your budget, not your optimism.
- **Grounds:** engine-to-frame and engine-to-body, new straps on clean metal. Half the 'bad PCM' stories in the swap world are a ground strap on paint.
- **The OBD-II port** goes where you can reach it with the door closed; future-you, at midnight, thanks present-you.

Chapter 6 — The Fuel System: 58 Pounds and Two Saddle Tanks

The LS wants a regulated 58 psi at the rail, EFI-rated line everywhere, and filtration (coarse before the pump, fine after). What the truck brings: saddle tanks that predate EFI by a decade and a half — unless yours is a 1987, and then it brings the swap's favorite starting point. Three cases, honestly:

- **1987 (TBI) trucks:** in-tank pumps already exist, at TBI pressure. Swap high-pressure pumps (the AC Delco EP381/EP241 class) into the 1987 sending units, keep the steel lines (they handle LS pressure), and regulate with the Corvette-style filter/regulator near the tank — about 58 psi to a returnless rail. The shortest road in the swap world.
- **Pre-'87 single tank:** the carbureted tank has no baffles and no sump; an inline pump sipping through the stock pickup will starve on corners at low fuel (documented, repeatedly). The done-once fix is an in-tank pump module or a baffled EFI tank — not a louder inline pump.
- **Dual saddle tanks:** the stock selector valve is a carburetor-pressure part that doesn't switch the return. The EFI-rated 6-port valve (the Pollak class, rated ~65 psi) switches feed AND return; two in-tank pumps, powered one at a time by the dash switch. Done this way the dual tanks work exactly like stock — from either side, at full pressure.



Note. Fire discipline, plainly: 58 psi finds weak hose the way water finds a roof leak. EFI-rated hose and proper clamps only — carburetor hose at LS pressure is a fire with a schedule. Prime the system and leak-check at pressure with eyes and nose before the first crank (Chapter 13), and re-check after the first heat cycle.

Chapter 7 — Cooling and the Steam Lines

The LS family's one non-negotiable: the steam vents at the top of the heads get plumbed to the pressurized side (radiator or surge point). Air trapped at the heads makes local steam, local hot spots, and a gauge that lies calmly while it happens. The truck bay is generous and the cores are big, which tempts builders to inherit the whole stock setup — inherit the core size if it's sound, but decide the fan: the mechanical clutch fan from a truck donor is honest and simple; electrics free the packaging and demand the shroud fit be measured, not vibes. Overheating diagnosis order (any swap, this one included): steam vents first, airflow second, capacity third, tune last.

Chapter 8 — Exhaust: Headers, Manifolds, and the Steering Box

The squarebody's constraint has a name: the steering box, inside the driver's frame rail, exactly where the driver's header wants to be. Swap-specific headers exist for this body (long-tube and mid-length lines built around that box); truck LS manifolds commonly clear everything — and per this family's standing honesty rule, manifolds lose some power and clear everything, and on a working truck that trade is often right. Pick with the box measured (Worksheet B), the engine at its final position, and the clutch or shift linkage in the picture if yours is a manual. Ground clearance: these trucks get lowered, and a long-tube collector is the first thing a speed bump interviews — the mid-lengths exist for a reason.

Chapter 9 — ECU and Tuning Decisions

Stock calibration with the swap edits (anti-theft out, fan outputs on, gear/tire math corrected, rear-O2 handling per your jurisdiction) runs a stock donor engine exactly like the donor truck did. A mail-order tune fixes the edit list and mild combinations; it cannot fix a cam it has never seen or fueling it cannot measure. A dyno tune earns its money on cams, forced induction, and combinations — and is sold as a habit on stock swaps. Decide by what changed, not by what the internet is doing this month.

Chapter 10 — The Gotchas

Each gotcha in the family's fixed format: *What happens* / *Why it happens* / *How to see it coming* / *The fix, in order of cost*. Categories the spec requires graded for this body are all here; where a category is clear on the squarebody, it says so in one line.

- **The carb-era selector valve at 58 psi.** *What:* the dual-tank truck loses fuel pressure, cross-fills tanks, or returns fuel to the wrong tank. *Why:* the stock valve was built for carburetor pressure and has no return switching. *See it:* Chapter 6, before the first prime. *Fix, in cost order:* single-tank the truck (honest, loses range); EFI-rated 6-port valve + two in-tank pumps (the factory-like answer).
- **Unbaffled tanks and the thirsty corner.** *What:* the truck stumbles or stalls on ramps and corners below a quarter tank. *Why:* pre-EFI tanks have no sump; the pickup breathes air. *See it:* any pre-'87 tank, by definition. *Fix:* in-tank module or baffled EFI tank; 'keep it above half' is a workaround, not a fix.
- **Steam pockets.** Covered in Chapter 7 — graded here: applies to every LS/LT swap; vent the heads, or the heads will explain it to you.



- **Header vs steering box.** *What:* the driver's-side header meets the box or the shaft. *Why:* both want the same rail-side inches. *See it:* Worksheet B at mock-up, box and shaft installed. *Fix:* swap-specific headers, or the manifolds that clear everything (see Chapter 8's honest trade).
- **The 4L80E vs the low-hump floor.** *What:* the big overdrive kisses the tunnel. *Why:* low-hump cabs were drawn around smaller cases. *See it:* Chapter 2's hump measurement against the transmission's published case dimensions. *Fix:* crossmember position first; floor relief last — it's permanent, so it's last.
- **Front driveshaft (4WD) vs the new crossmember.** *What:* the shaft meets the moved crossmember at droop. *Why:* the transmission moved, the axle didn't. *See it:* mock-up at full droop, not at ride height. *Fix:* crossmember selection; transfer-case clocking only with the driveline angles re-measured (Worksheet A).
- **Throttle-pedal geometry (drive-by-wire).** *What:* a DBW pedal at the wrong angle makes a lazy or touchy truck forever. *Why:* the PCM reads angle, not effort. *See it:* mock the mount; verify full travel against the stop. *Fix:* reposition; no tune cures geometry.
- **Donor accessory completeness.** *What:* the 'complete pullout' lacks the pedal, MAF, or PCM. *Why:* 'complete' is a sales word. *See it:* Chapter 2's signed checklist. *Fix:* inside the negotiation, never after it.
- **Driveshaft angle after a drop.** *What:* vibration that survives balancing. *Why:* the lowering changed the working angles. *See it:* Worksheet A at final height. *Fix:* angle correction per the drivetrain guides; a shaft built to your measured length.
- **Checked — clear on this body:** firewall and inner-fender clearance (the bay is a truck bay); shock towers (there are none); A/C evaporator vs headers is a routing check, not a structural one, in this bay.

Chapter 11 — Tips and Tricks

- **Plan on three engine installations:** mock-up, clearance-check, final. The builders who do it once planned on three and got lucky twice.
- **Set the engine's final fore-aft position with the steering, fan, and transmission all in the picture** — a position that pleases one will be re-litigated by the other two.
- **Drop the tanks once, do everything:** pumps, senders, vent lines, and the valve, while access is free. Tank work twice is a choice.
- **Bag and tag the donor teardown;** the connector you cut 'because it looked unused' is the one the PCM wants.
- **Photograph every routing decision;** heat and chafe are patient, and the photo is the map at the first repair.
- **Chase threads before torque values** — 40-year-old frame threads lie to torque wrenches.

Chapter 12 — Special Tools

Must-have: a hoist with an engine leveler (the leveler matters more than the hoist upgrade — the engine goes in at an angle, over a core support), stands you trust, a fuel-pressure gauge that reads over 60 psi (Chapter 6's verification is not optional), a multimeter, thread chasers for the frame's sizes. **Rent or borrow:** the transmission jack (a 4L80E on a floor jack is a hospital story), the balancer puller/installer, a flare tool if hard lines are being bent. **Nice:** a lift — it turns Chapter 13's under-truck checks from an afternoon into an hour. No tool is listed without the chapter that uses it.



Chapter 13 — First Start and Shakedown

- **Before cranking:** fuel system primed and leak-checked at full pressure (both tanks, both selector positions, if yours is dual); vents verified; oil primed per the engine's procedure; PCM security status confirmed; extinguisher in reach.
- **First start:** one person at the key, one at the bay with a light. Short cranks; oil pressure is the only gauge today.
- **First heat cycle:** fan trigger verified against a real temperature; steam vents re-checked; tank selector exercised warm, at idle, while someone watches for leaks.
- **50 miles:** leaks, belts, harness chafe, selector behavior on the road, driveline report (Worksheet A if it vibrates). **500 miles:** re-torque list, fluid levels, alignment after springs settle — and the honest Chapter 9 conversation about what the tune still owes you.

Note. Safety, plainly: verify the brakes at walking pace, then neighborhood pace, before any road speed — the swapped truck has different weight over the front axle and a different pedal story. Steering-box fasteners torque into inspected metal, per the factory literature — never 'snug by feel' on the part that aims the truck.

Appendix: Worksheets (Fill These In On YOUR Car)

Worksheet A — Driveline. Ride height (four corners, as driven): _____. Transmission output angle: _____. Pinion angle: _____. Working angles computed: _____. Shaft length, center-to-center at ride height: _____. 4WD: front shaft clearance at full droop: _____.

Worksheet B — Tunnel and clearance. Cab floor: high-hump / low-hump (measured): _____. Transmission chosen: _____. Bellhousing-to-floor clearance: _____. Crossmember: _____. Header-to-steering-box clearance: _____. Pan-to-crossmember clearance, lock to lock: ____ / ____.

Worksheet C — Parts-total ledger. Engine/donor: _____. Mounts/crossmember: _____. Fuel system (pumps, valve, lines, regulator): _____. Cooling: _____. Harness/PCM/tune: _____. Exhaust: _____. Transmission and driveshaft: _____. Gauges and small parts: _____. Total, honest: _____. Compare it to the crate quote and to finishing the truck as-is before Chapter 1's decision is final — that comparison IS the decision.

Recommended for this restoration

Shop exact-fit supplies on Amazon

- LS swap squarebody 1973-1987 motor mount kit
- 1987 squarebody EFI fuel tank sending unit LS swap
- LS swap Corvette fuel filter regulator 58 psi
- squarebody LS swap headers

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Sources Consulted

Every specification in this guide was checked against published references before printing. Where sources disagree, this guide says so in the text rather than picking a number silently. The references below are the backbone of the technical chapters:

- **Manufacturer fitment documentation (ICT Billet and peers).** squarebody adapter-plate mount kits — the published crank-centerline/bellhousing-position claims and the low-mount compressor fitment notes
- **Published LS-swap fuel-system references, 1973–87 C/K.** the 1987 TBI in-tank-pump conversion path (high-pressure pump swap in the 1987 sender), the Corvette-style filter/regulator at ~58 psi, baffled vs unbaffled tank behavior, and the EFI-rated 6-port selector-valve solution for dual tanks
- **GM LS-family production data.** engine RPO applications and factory ratings for Gen III, Gen IV, and Gen V engines as published in GM engine-family references
- **GM C/K platform documentation.** 1973–87 platform history, saddle-tank and transfer-case applications, and the R/V-series continuation
- **EFI system manufacturers' installation literature.** fuel-pressure and filtration requirements as published

A note on numbers. Factory literature, period road tests, and marque historians do not always agree — production totals, horsepower ratings, and paint codes all have documented conflicts. This guide uses the best-supported figure in each case and flags the conflict where one exists, so you can judge for yourself.

Emissions legality varies by jurisdiction — these trucks straddle emissions-era lines in many states: this guide explains what gets asked; it cannot guarantee any conversion passes anywhere. Structural rust and frame repair belong to the restoration guides and to shops; forced induction beyond the factory-supercharged engines is outside this guide's scope.



Thank You

Thank you for choosing this RestorationEssentials guide. Whether this is your first LS/LT Conversion — GM Squarebody 1973–1987 or your fiftieth, you picked one of the great ones, and the fact that you are doing your homework before you turn a wrench puts you ahead of most of the field.

These cars survived because people like you refused to let them disappear. Every correct fastener, every properly decoded tag, every hour spent getting the details right keeps the history honest for the next owner and the next generation of enthusiasts.

Modern engine, working truck — planned before the first wrench turned. Measure on your truck, write it in the worksheets, and finish it: a finished swap beats a perfect plan every Saturday night.

If this guide helped your build, we would love to hear about it. Good luck with the restoration, and enjoy every mile.

— The RestorationEssentials Team